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A Synopsis on

"INTELLIGENT CAREER PATH ADVISORY AGENT FOR SENIOR IT PROFESSIONALS"

Submitted for the Major Project Work-Phase 1
(Academic Year 2025-26)

Submitted by

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Declaration by Students

We, the undersigned students **Deepanshu Bisht (1NT23CS056) and Divya S Karki (1NT23CS067)** hereby declare that we are fully aware of the **project conduction procedures** and **evaluation rubrics** as prescribed by the department. We understand that the **project work will be evaluated through continuous assessment** across the defined phases.

Name of the Student	***Signature***
Deepanshu Bisht	
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Confirmation by Guide

I, **Dr. Krishna Rao Venkatesh**, hereby declare that I shall guide the above-mentioned students in the project entitled **"INTELLIGENT CAREER PATH ADVISORY AGENT FOR SENIOR IT PROFESSIONALS"** and will ensure that the project work is carried out under my supervision and submitted as per the academic requirements for the academic year **2025–2026**.

Signature of the Guide

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1. Introduction

In today's rapidly evolving technology landscape, experienced IT professionals face career decisions that are fundamentally more complex than those encountered at the start of their careers. A senior engineer with five or more years of experience must navigate choices such as deepening a technical specialization, transitioning into people management, pivoting to an adjacent domain like product management or AI, or preparing for an executive role such as Engineering Director or Chief Technology Officer. These are not questions that a job portal or a keyword-based resume matcher can answer.

Existing career platforms such as LinkedIn, Naukri, and Glassdoor were designed primarily for job matching. They focus on aligning qualifications with job descriptions and are optimized for early-career hiring. For senior professionals, these platforms provide limited value because they do not assess leadership readiness, Emotional Intelligence (EQ), mentoring capability, team collaboration patterns, or the nuanced skill gaps that separate a strong individual contributor from a high-performing leader. Moreover, they offer no structured guidance for career transitions, no mentor matching based on shared professional history, and no mechanism for long-term goal tracking.

This project proposes the development of an Intelligent Career Path Advisory Agent for Senior IT Professionals, a sophisticated AI-powered system that addresses the specific and underserved advisory needs of experienced technology practitioners. The system leverages Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), multi-agent reasoning, and a five-dimensional psychometric assessment engine that explicitly includes Emotional Intelligence evaluation. It goes beyond recommendation to deliver personalized leadership transition roadmaps, algorithmic mentor matching, SMART goal tracking, and real-time industry intelligence.

By integrating psychological assessment, skill gap analysis, executive coaching support, and agentic AI planning into a single unified platform, this project aims to transform how senior IT professionals navigate the most consequential decisions of their careers. The system is explicitly scoped to experienced professionals and is designed to avoid overlap with general-purpose placement agents that target students and early-career job seekers.

2. Objectives of the Project

The following specific and measurable objectives guide the development of the Intelligent Career Path Advisory Agent for Senior IT Professionals:

1. To design and implement a five-dimensional psychometric profiling engine that assesses personality, career motivators, learning styles, technical skills, and Emotional Intelligence (EQ), providing a comprehensive model of the senior professional's leadership readiness and career alignment.
2. To build an AI-powered skill gap analysis module that compares a professional's current competencies against verified industry benchmarks for senior roles such as Technical Architect, Engineering Manager, and Chief Technology Officer, and generates a structured, prioritized learning roadmap.
3. To develop an LLM-powered agentic career path recommendation engine capable of modeling high-level role transitions such as Senior Engineer to Engineering Manager and Engineering Manager to CTO, with explainable reasoning for each suggested path.
4. To create an algorithmic mentor matching system that connects users with experienced IT leaders who have undergone comparable career transitions, using vector similarity over structured transition profiles.
5. To integrate SMART goal tracking with milestone management and succession planning visibility, enabling professionals and enterprise users to monitor progress toward defined career objectives over time.
6. To incorporate real-time labor market intelligence into the advisory pipeline, ensuring that career recommendations are grounded in current industry demands, emerging skill trends, and role availability data.

3. Literature Survey and Review

A comprehensive review of existing literature was conducted across the areas of career recommendation systems, intelligent advisory agents, leadership development frameworks, Emotional Intelligence in professional settings, and the application of large language models in personalized guidance. The following summarizes the key findings and their relevance to this project.

Early career recommendation systems relied heavily on collaborative filtering and content-based filtering techniques, similar to those used in e-commerce recommendation engines. These approaches matched users to job profiles based on keyword overlap in resumes and job descriptions. While effective for simple matching, such systems lacked the ability to reason over complex user traits like personality, leadership potential, soft skills, or aspirational career transitions. Studies by Shalaby et al. (2017) and Qin et al. (2018) highlighted that purely skill-keyword-based matching results in low precision for non-linear career paths and career changers, a limitation that is even more pronounced for senior professionals navigating executive transitions.

The foundational work on Emotional Intelligence by Salovey and Mayer (1990), later popularized by Goleman (1995), established that EQ competencies including self-awareness, empathy, and relationship management are strong predictors of leadership effectiveness. Subsequent research by Boyatzis (2008) demonstrated that EQ is a more reliable differentiator of high-performing leaders than technical expertise alone. This motivates the explicit inclusion of an EQ assessment module in the proposed system, particularly given that the target users are professionals seeking leadership roles.

Research in personality-aware recommendation systems, particularly those incorporating the Big Five personality model (OCEAN), demonstrated that integrating psychological traits into career matching significantly improves user satisfaction and long-term career alignment. Park et al. (2021) showed that personality-augmented recommendation engines outperformed skill-only systems in predicting job satisfaction outcomes, reinforcing the need for multi-dimensional profiling beyond technical qualifications.

The emergence of transformer-based language models such as BERT and GPT marked a significant shift in how career data could be processed and understood. Bian et al. (2020) demonstrated that contextual embeddings from pretrained models significantly improved job-skill alignment tasks compared to traditional TF-IDF or Word2Vec approaches. The concept of autonomous agents built on top of LLMs gained traction following the ReAct framework (Yao et al., 2022). These demonstrated that LLMs could iteratively plan, reason, and execute multi-step tasks by interacting with external tools and knowledge

bases, a paradigm directly applicable to career advising where an agent must gather profile data, retrieve industry benchmarks, identify gaps, and synthesize a structured roadmap.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), strengthened the ability of LLMs to answer domain-specific queries by grounding responses in retrieved external knowledge. In the context of senior career guidance, RAG enables the system to retrieve current leadership role requirements, salary benchmarks, and emerging technology demands before generating recommendations, mitigating hallucination and ensuring factual accuracy.

Overall, the literature reveals that while individual components such as EQ assessment, LLMs, RAG, and agentic planning have been studied in isolation, no existing system comprehensively integrates all these elements into a unified advisory platform for senior IT professionals. This gap forms the primary motivation for the proposed project.

4. Limitations of the Existing Work (Gap Analysis) and Motivation

A critical review of existing career guidance systems and related research reveals several significant limitations that the proposed Intelligent Career Path Advisory Agent for Senior IT Professionals aims to address.

4.1 Platforms Are Designed for Early-Career Users, Not Senior Professionals

Mainstream career platforms such as LinkedIn, Naukri, and Glassdoor are optimized for resume-to-job-description matching and are primarily used by students and early-career job seekers. They do not support the advisory reasoning required by senior professionals, such as modeling a transition from Individual Contributor to Engineering Manager, assessing readiness for a CTO role, or identifying mentors with relevant transition experience. The needs of a Senior Software Engineer contemplating a move into Engineering Leadership are fundamentally different from those of a fresher seeking their first job.

4.2 Absence of Emotional Intelligence and Leadership Readiness Assessment

Existing career platforms assess technical skills and academic qualifications but entirely neglect Emotional Intelligence (EQ), which is widely recognized as a critical predictor of leadership success. Senior professionals aiming for management or executive roles require feedback on their self-awareness, empathy, conflict resolution, and team influence capabilities. The absence of EQ evaluation means professionals may receive technically accurate career suggestions that are misaligned with their actual leadership maturity.

4.3 No Modeling of Leadership Transition Pathways

Current systems can suggest job roles based on current skills but are unable to model multi-step career transitions such as Senior Engineer to Engineering Manager, or Technical Lead to Chief Technology Officer. These transitions involve not only skill acquisition but also shifts in mindset, influence patterns, and organizational behavior. No existing platform provides structured guidance for navigating such transitions with explanations and sequenced milestones.

4.4 No Mentor Matching Based on Transition History

While some platforms offer generic networking features, none provide algorithmic mentor matching based on transition similarity. A senior professional planning to move from an individual contributor role into management would benefit significantly from being connected with a mentor who has made the same transition in a comparable domain. Existing tools do not support this kind of structured, similarity-driven mentorship pairing.

4.5 Static Guidance Without SMART Goal Tracking or Succession Planning

Career guidance in current platforms is typically delivered as a one-time assessment or a static list of suggestions. There is no mechanism for setting structured SMART goals with milestones, tracking progress over time, or surfacing internal succession planning opportunities within an enterprise. This makes existing tools unsuitable for the sustained, multi-phase career development that senior professionals require.

4.6 Motivation

The limitations identified above collectively underscore the need for an intelligent, explainable, and senior-focused career advisory system that goes beyond job matching to address the complex, multi-dimensional nature of experienced IT professionals' career decisions. The Intelligent Career Path Advisory Agent for Senior IT Professionals is motivated by the opportunity to bring together LLMs, agentic planning, RAG, EQ-inclusive psychometric assessment, and algorithmic mentor matching into a unified platform that genuinely serves this underserved population.

5. Functional Requirements for the Prototype (Major Project-Phase 1)

The prototype of the Intelligent Career Path Advisory Agent for Senior IT Professionals focuses on demonstrating the core feasibility of the system. The following functional requirements define the key capabilities that the prototype must deliver within the scope of Major Project Phase 1.

Req ID	Functional Requirement (SMART Compliant)	SMART Justification	Expected Outcome
FR1	The system shall allow a senior IT professional to complete a structured five-dimensional assessment covering technical skills, soft skills, career motivators, personality traits, and Emotional Intelligence (EQ), and store the results as a structured user profile.	**Specific:** Five-dimensional assessment including EQ. **Measurable:** Profile stored and retrievable. **Achievable:** Form-based interface with EQ questionnaire. **Relevant:** Core input for senior-level advisory. **Time-Bound:** Implemented in Sem VI prototype.	A complete multi-dimensional profile is created and used as input to the career advisory and skill gap analysis engine.
FR2	The system shall analyze the user's profile using an LLM-powered reasoning engine and generate at least three personalized senior-level career path recommendations with explanations, including at least one leadership transition pathway such as Senior Engineer to Engineering Manager.	**Specific:** LLM-based senior role recommendations. **Measurable:** At least 3 paths with reasoning including one leadership transition. **Achievable:** Using Gemini/GPT-4o API with LangChain. **Relevant:** Core advisory function for senior users. **Time-Bound:** Delivered within prototype scope.	User receives personalized, explained career recommendations specific to their experience level, not generic job suggestions.
FR3	The system shall perform a skill gap analysis by comparing the user's current competency profile against verified requirements for each recommended senior role and present a structured, prioritized learning roadmap with specific resources for each identified gap.	**Specific:** Gap analysis against senior role benchmarks. **Measurable:** Gaps identified per recommended role with resources. **Achievable:** LLM + RAG pipeline with curated knowledge base. **Relevant:** Bridges user profile to senior career goals. **Time-Bound:** Implemented in Sem VI prototype.	User receives a structured, actionable plan showing which skills to acquire, in what priority order, and through what specific learning resources.

6. Proposed Methodology for Building Prototype (Major Project-Phase 1)

The prototype will be developed following a structured, modular methodology that integrates five-dimensional psychometric profiling including EQ assessment, LLM-based agentic reasoning, retrieval-augmented generation, skill gap analysis, and a conversational interface. The overall workflow is designed to demonstrate core feasibility within the scope of Major Project Phase 1.

6.1 System Architecture

The system is organized into the following core modules:

- User Profiling and EQ Assessment Module: Collects multi-dimensional user data through a structured onboarding interface, including a dedicated Emotional Intelligence questionnaire and SWOT self-evaluation tool.
- LLM Reasoning Engine: Processes the structured user profile using a large language model to generate senior-level career path recommendations with explainable reasoning.
- RAG Pipeline: Retrieves current senior role benchmarks, leadership competency frameworks, and industry trend data from a curated vector knowledge base to ground recommendations in real-world data.
- Skill Gap Analyzer: Compares the user's current competency profile against retrieved role requirements and generates a prioritized learning roadmap with specific resources.
- Mentor Matching Agent: Uses vector similarity over structured transition profiles to surface compatible mentors who have made transitions similar to the user's target path.
- Conversational Interface: Provides a natural language chat interface for users to interact with the advisory agent, ask follow-up questions, and refine recommendations.

6.2 Workflow

The end-to-end data and reasoning flow of the system is as follows:

User Onboarding + EQ Assessment → Structured Profile Storage → RAG-Augmented Role Benchmark Retrieval → LLM Agentic Reasoning → Career Path Recommendations + Skill Gap Analysis → Mentor Matching → Conversational Output with Goal Tracking

6.3 Software and Tools

Component	Technology / Tool
Programming Language	Python 3.11
LLM API	Google Gemini API / OpenAI GPT-4o
Agentic Framework	LangChain / LangGraph
RAG & Vector Store	FAISS / ChromaDB with LangChain
Backend API	FastAPI
Frontend Interface	React.js
Database	SQLite / Firebase (user profiles)
Version Control	Git and GitHub

6.4 Algorithms and Techniques

- Retrieval-Augmented Generation (RAG): Used to retrieve current senior role requirements, leadership competency benchmarks, and industry insights from a curated vector knowledge base before LLM inference.
- Agentic Chain-of-Thought Reasoning: The LLM agent decomposes career advisory queries into sub-tasks, reasons step-by-step over the user's profile and retrieved data, and synthesizes a coherent, explainable response.
- Semantic Similarity Matching: User competency profiles and mentor transition histories are encoded as vector embeddings and compared using cosine similarity to identify skill gaps and compatible mentors.
- Psychometric and EQ Scoring: Standardized questionnaires generate structured scores across the five assessment dimensions, with the EQ component drawing on the Mayer-Salovey-Caruso model.
- Prompt Engineering: Carefully designed system prompts guide the LLM to produce structured, senior-role-appropriate career roadmaps tailored to the user's experience level and target transition.

6.5 Development Approach

The prototype will be built iteratively. In the first phase, the user onboarding interface and EQ-inclusive profiling module will be implemented and tested. In the second phase, the RAG pipeline will be integrated to ground recommendations in real-time senior role data. Finally, the skill gap analyzer, mentor matching module, and conversational interface will be completed, and end-to-end testing will be performed to validate all three functional requirements.

7. GitHub Repository Details

The project source code, datasets, experiment notebooks, and documentation are maintained in a GitHub repository to ensure transparency, version control, and collaborative development. The repository will be updated progressively throughout the development lifecycle.

The project repository is available at:

https://github.com/nmit-1NT23CS056/final_year_project

The repository will include the implementation of all modules including the EQ assessment interface, LLM reasoning engine, RAG pipeline, skill gap analyzer, mentor matching module, and the conversational frontend. Supporting resources such as sample user profiles, vector knowledge base setup scripts, and API integration guides will also be included to ensure reproducibility.

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9. Annexures

9.1 Annexure-1: Definition of Prototype or Proof-Of-Concept

The prototype developed in this project demonstrates the core feasibility of an LLM-powered Intelligent Career Path Advisory Agent specifically designed for senior IT professionals. The aim of the prototype is to validate that a multi-dimensional user profile, including EQ scores and a SWOT self-evaluation, can be processed by an agentic AI system to generate personalized, explainable, and actionable career recommendations suited to experienced professionals.

The prototype accepts structured user inputs covering technical skills, soft skills, career motivators, personality traits, and Emotional Intelligence. These inputs are processed by the LLM reasoning engine, augmented by a RAG pipeline that retrieves current senior role benchmarks and leadership competency frameworks. The system generates career path recommendations, including at least one leadership transition pathway, accompanied by clear reasoning and a structured skill gap analysis pointing to specific learning resources.

While the prototype does not aim to achieve production-level robustness or cover all career domains and executive roles, it successfully demonstrates the technical feasibility and practical value of combining five-dimensional psychometric profiling, LLM agentic planning, and RAG for senior IT career advisory purposes. The insights gained will directly inform the final system architecture and feature set in the subsequent semester.

9.2 Annexure-2: Definition of Functional Requirements

Functional Requirements describe what the system must do. They define the behaviors, features, and capabilities that the prototype must demonstrate. In this project, functional requirements are written to be SMART compliant, meaning each requirement is Specific, Measurable, Achievable, Relevant, and Time-Bound.

The following table provides the complete SMART-justified functional requirements for the prototype:

Req ID	Functional Requirement (SMART)	SMART Justification	Expected Outcome
FR1	The system shall allow a user to complete a five-dimensional assessment including EQ and store results as a structured profile.	**Specific:** Five-dimension profile capture. **Measurable:** Data stored and retrievable.	A complete multi-dimensional profile is created and used as input to the advisory engine.

		Achievable: Form UI with EQ module. **Relevant:** Core senior advisory input. **Time-Bound:** Sem VI.	
FR2	The system shall generate at least three senior-level career path recommendations with explanations, including at least one leadership transition pathway.	**Specific:** LLM senior role recommendations. **Measurable:** 3+ paths with reasoning. **Achievable:** Gemini/GPT API. **Relevant:** Core advisory function. **Time-Bound:** Prototype scope.	User receives senior-appropriate, explained career recommendations including leadership transitions.
FR3	The system shall identify skill gaps against senior role benchmarks and present a structured, prioritized learning roadmap with resources.	**Specific:** Gap analysis vs senior benchmarks. **Measurable:** Gaps per role with resources. **Achievable:** LLM + RAG pipeline. **Relevant:** Actionable senior guidance. **Time-Bound:** Sem VI.	User receives a structured plan showing which skills to acquire and in what order to reach target senior roles.